Technical Appendix: Annotation-Informed Priors

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This appendix describes how `sblr` transfers marker annotations into sparse Bayesian linear-regression priors. It develops marker-specific inclusion probabilities and effect-variance scales, explains the fixed, learned, and group-prior CSR wrappers, and records empirical-Bayes constructions from the original annotation-prior draft. The emphasis is on the prior and its implementation contract; the CSR likelihood and sparse-LD representation are covered elsewhere.

1 Annotation Matrix and Preprocessing

Let

$$A = \begin{bmatrix} A_{11} & \cdots & A_{1K} \\ \vdots & \ddots & \vdots \\ A_{m1} & \cdots & A_{mK} \end{bmatrix}$$

be an $m \times K$ marker-by-annotation matrix. Row A_i describes marker i, and column k describes an annotation. Binary annotations may encode membership in functional sets; quantitative annotations may encode continuous scores. A marker may belong to several annotations, so the columns generally do not define a partition.

The marker-row contract is strict. Rows of A must correspond, in the same order, to:

- the markers used to compute `wy` and `ww`;
- rows and columns of the sparse-LD matrix;
- marker names represented by `cls` and `Glist$rsidsLD`; and
- rows of posterior marker summaries such as `bm` and `dm`.

When row names and marker names are both available, the annotation preparation helper can reorder rows by name. Names are not a substitute for checking the alignment: duplicated, incomplete, or stale names can still produce an incorrect analysis.

1.1 Intercepts, binary columns, and quantitative columns

The annotation-aware wrappers use `.stblr_prepare_annotation_matrix()` to validate and preprocess A. Its relevant behavior is:

- values must be numeric and finite;
- an intercept is recognized when the first column is all ones;
- `add_intercept = TRUE` prepends an intercept if one is not recognized;
- with `standardize = TRUE`, nonbinary, non-intercept columns are centered and scaled;
- binary columns are left unchanged unless `center_binary = TRUE`.

An intercept lets the model express a baseline prior independently of the annotation contrasts. Centering quantitative columns makes their coefficients more comparable and reduces confounding between the intercept and annotation effects. Leaving a binary indicator uncentered gives its coefficient the interpretation of a contrast against markers outside that annotation.

These choices are part of the model specification and should be reported. Automatic standardization is convenient, but users should retain the preprocessing metadata needed to apply the same transformation in sensitivity analyses or prediction data.

2 Baseline Sparse Prior

For a single trait, the baseline sparse prior introduces a marker inclusion indicator d_i:

$$
\begin{aligned}
b_i \mid d_i = 0 &= 0, \\
b_i \mid d_i = 1 &\sim N(0, v_b), \\
\Pr(d_i = 1) &= \pi.
\end{aligned}
$$

Here b_i is the effect of marker i, v_b is the active-marker variance, and π is a global inclusion probability. The posterior inclusion probability (PIP) is

$$\Pr(d_i = 1 \mid \text{data}),$$

which is summarized by `dm` when returned by a fitted model.

The annotation-informed models retain this sparse-prior logic but replace one or both global quantities with structured marker-specific quantities:

$$\Pr(d_i = 1) = \pi_i, \qquad b_i \mid d_i = 1 \sim N(0, v_b w_i),$$

where $w_i > 0$ is a marker-specific variance multiplier.

3 Marker-Specific Prior Construction

3.1 Inclusion probabilities

A common annotation construction places annotation effects on the log-odds scale:

$$\text{logit}(\pi_i) = \text{logit}(\pi_0) + A_i \eta_\pi,$$

where π_0 is a baseline inclusion probability and η_π contains annotation coefficients. Positive values of $A_i \eta_\pi$ increase prior inclusion odds; negative values reduce them.

The fixed-prior helper used by the wrappers centers the linear predictor before combining it with the baseline:

$$\tilde{\ell}_{\pi i} = A_i \eta_\pi - \frac{1}{m} \sum_{j=1}^{m} A_j \eta_\pi, \qquad \pi_i = \text{logit}^{-1} \left\{ \text{logit}(\pi_0) + \tilde{\ell}_{\pi i} \right\}.$$

Centering makes the annotation effects primarily redistribute prior inclusion probability across markers rather than changing its genome-wide location. The resulting values are clamped to user-specified lower and upper bounds for numerical and inferential stability.

3.2 Effect-variance multipliers

Annotation effects on active-marker variance naturally use a log scale:

$$\log w_i = A_i \eta_v, \qquad w_i = \exp(A_i \eta_v).$$

As with inclusion probabilities, the wrapper centers the linear predictor:

$$\tilde{\ell}_{vi} = A_i \eta_v - \frac{1}{m} \sum_{j=1}^{m} A_j \eta_v, \qquad w_i = \exp(\tilde{\ell}_{vi}).$$

This construction makes w_i a relative multiplier around the global active-marker variance v_b. Lower and upper bounds prevent extreme annotation combinations from creating nearly degenerate or unreasonably diffuse marker priors.

3.3 Consequences for a single-marker update

The marker-specific quantities enter the standard sparse BLR update directly. Let w_i^{XX} denote the diagonal marker cross-product and let

$$s_i = r_i + w_i^{XX} b_i$$

be the marker score formed from the current residual state. To avoid confusing the cross-product with the prior multiplier, write $v_{b,i} = v_b w_i$. The log Bayes factor for the active versus null component is

$$\log BF_i = \frac{1}{2} \log \left(\frac{v_e}{v_e + w_i^{XX} v_{b,i}} \right) + \frac{1}{2} \frac{s_i^2 v_{b,i}}{v_e (v_e + w_i^{XX} v_{b,i})}.$$

The conditional inclusion probability is

$$\Pr(d_i = 1 \mid \cdot) = \frac{\pi_i BF_i}{(1 - \pi_i) + \pi_i BF_i}.$$

Conditional on inclusion,

$$b_i \mid d_i = 1, \cdot \sim N \left(\frac{s_i}{w_i^{XX} + v_e / v_{b,i}}, \frac{v_e}{w_i^{XX} + v_e / v_{b,i}} \right).$$

Thus, π_i alters prior odds of inclusion, whereas w_i alters the active effect distribution and its shrinkage. These mechanisms should not be interpreted as interchangeable.

4 Annotation-Level Posterior Summaries

Posterior marker summaries can be aggregated to describe signal associated with an annotation set $S_k = \{i : A_{ik} \neq 0\}$. Useful descriptive summaries include:

$$m_{\mathrm{eff},k} = \sum_{i \in S_k} \widehat{\Pr}(d_i = 1 \mid \mathrm{data}),$$

the expected number of included markers in the set, and the PIP enrichment ratio

$$E_k = \frac{|S_k|^{-1} \sum_{i \in S_k} \widehat{\Pr}(d_i = 1 \mid \mathrm{data})}{|S_k^c|^{-1} \sum_{i \notin S_k} \widehat{\Pr}(d_i = 1 \mid \mathrm{data})}.$$

For summary-statistics fits, a marker-level contribution estimate may be written

$$c_i = \frac{\hat{b}_i \{wy_i - r_i\}}{n},$$

with an optional PIP-weighted version. An annotation-level genetic-variance summary is then

$$\widehat{V}_{G,k} = \sum_{i \in S_k} c_i.$$

These set summaries are deliberately descriptive. When annotations overlap, their totals are not additive and the same marker contribution may appear in several sets. `summarize_annotation_signal()` follows this descriptive set-based approach for known simulation truth and, when a fit is supplied, adds within-set means of `dm` and absolute `bm`.

For conditional interpretation with overlapping annotations, one can regress a marker-level posterior response s_i on all annotation columns:

$$s_i = \alpha + \sum_{k=1}^{K} A_{ik} \gamma_k + \epsilon_i.$$

Possible responses include a PIP, squared posterior mean effect, or estimated marker contribution. Such a regression estimates conditional associations among annotations, but it does not establish that an annotation causes the genetic signal.

5 Empirical-Bayes Prior Construction

An empirical-Bayes workflow estimates annotation signal from one genomic fit and uses it to construct priors for a subsequent fit:

$$\text{initial genomic fit} \to \text{marker summaries} \to \text{annotation model} \to \{\pi_i, w_i\} \to \text{updated genomic fit}.$$

This approach separates prior construction from the genomic sampler. It is useful when annotation effects should be estimated with a dedicated model or when fixed marker priors must be inspected before fitting.

5.1 Set-based estimates

A smoothed set-level inclusion probability can be estimated from PIPs:

$$\hat{\pi}_k = \frac{a + \sum_{i \in S_k} dm_i}{2a + |S_k|},$$

where $a > 0$ supplies symmetric pseudo-count shrinkage. A rough active-effect variance estimate is

$$\hat{v}_{b,k} = \frac{\widehat{V}_{G,k}}{\max(m_{\text{eff},k}, c)},$$

where $c > 0$ prevents instability when the expected number of active markers is very small. Both estimates should generally be shrunk toward genome-wide values before they are converted into marker priors.

These formulas are easiest to interpret for disjoint sets. With overlapping annotations, directly assigning each set estimate to its member markers can double-count evidence.

5.2 Conditional annotation models

For overlapping annotations, an annotation-level regression or BLR can map marker summaries to annotation coefficients. Candidate marker-level responses from the archived design include:

- posterior mean effects, $\hat{b}_i$;
- transformed PIPs, such as a bounded $\text{logit}(dm_i)$;
- squared effects or absolute effects; and
- marker contribution estimates, $\hat{b}_i(wy_i - r_i)/n$.

The fitted annotation effects can then be transformed through the centered logit and log-scale constructions above. Regularization is essential because annotation columns can be numerous, correlated, imbalanced, or nearly redundant.

5.3 Data reuse and cross-fitting

When the same phenotype data are used both to estimate annotation enrichment and to fit the annotation-informed model, the resulting prior is data-dependent. This can improve adaptation but can also exaggerate evidence if uncertainty in the first-stage estimate is ignored.

Useful safeguards include:

- shrink annotation coefficients toward zero;
- clamp π_i and w_i to defensible ranges;
- compare against an unannotated baseline;
- split markers into approximately LD-independent folds and construct priors out of fold; and
- assess sensitivity to the annotation model and prior bounds.

The current fixed-prior wrapper supports the final marker-prior fitting stage. Cross-fitting and a full empirical-Bayes pipeline remain analysis-level responsibilities rather than a single exported package workflow.

6 Fixed Marker-Specific Prior Wrapper

`stblr_csr_annot(annotation_model = "prior")` fits a CSR summary-statistics model with fixed marker-specific inclusion probabilities and effect-variance multipliers. Users may either supply `fixed_pi_marker` and `fixed_vb_multiplier` directly, or supply A and fixed annotation coefficients from which the wrapper constructs them. The compatibility wrapper `stblr_csr_prior_annot()` exposes the same fixed-prior backend.

```
fit_fixed <- stblr_csr_annot(
  stats = stats,
  Glist = Glist,
  annotations = list(
    A = A,
    beta_pi = beta_pi,
    beta_vb = beta_vb
  ),
  annotation_model = "prior"
)
```

The annotation-derived marker priors remain fixed during sampling. Marker effects, inclusion states, residual variance, global variance parameters, and other sampler states may still be updated according to the wrapper's update flags. Fixed therefore refers to the annotation-to-marker prior mapping, not to the whole fitted model.

This wrapper is appropriate when priors come from external evidence, a separate training analysis, or a deliberately staged empirical-Bayes procedure.

7 Learned Annotation Effects

stblr_csr_learn_annot() learns annotation coefficients jointly with the marker model. For trait t, the implemented construction is conceptually

$$\text{logit}(\pi_{it}) = \text{logit}(\pi_t) + \text{center}(A_i \eta_{\pi,t}),$$

and

$$v_{b,it} = v_{b,t} \exp\{\text{center}(A_i \eta_{v,t})\}.$$

The annotation effects are initialized by eta_pi_init and eta_vb_init, with zero matrices used by default. Gaussian shrinkage scales sigma_eta_pi and sigma_eta_vb regularize the coefficients.

The native sampler uses random-walk Metropolis-Hastings updates at the annot_update_every schedule. The inclusion-coefficient target is informed by the current inclusion states; the variance-coefficient target is informed by the current active effects and inclusion states. Proposal scales are controlled by rw_sd_eta_pi and rw_sd_eta_vb. Marker-level inclusion probabilities and variance multipliers are bounded during construction.

This is a direct learned-prior model rather than a separate two-stage empirical-Bayes fit. The posterior annotation coefficients summarize how the model adapts prior inclusion and active-effect variance, conditional on all other annotations and the sampler specification.

8 Group Priors

stblr_csr_group_annot() assigns each marker to exactly one group through group_index. Group g receives a group-specific inclusion probability and variance multiplier:

$$\Pr(d_i = 1) = \pi_{g(i)}, \qquad b_i \mid d_i = 1 \sim N\{0, v_b w_{g(i)}\}.$$

Group inclusion probabilities have beta-prior shape parameters derived from a prior mean and strength. Group variance multipliers may also be updated, depending on updateGroupVb, and can be normalized around a common scale.

The group construction is a simplification of overlapping annotation models: it creates an unambiguous partition and directly pools evidence within groups, but it cannot represent a marker's simultaneous membership in several functional categories. It is useful when annotations naturally define mutually exclusive strata or when a simpler, more stable model is preferred.

The preparation helper verifies that every marker has one nonmissing group, that requested group names define the ordering, and that every represented group is nonempty. The native interface receives a zero-based group index; users should supply ordinary R group labels and let the wrapper handle that conversion.

9 Simulation Helpers and Known-Truth Diagnostics

The annotation simulation helpers support controlled checks of whether an annotation-informed model recovers known architecture.

`make_overlapping_annotations()` creates binary annotation columns using independent membership probabilities. Overlap is allowed, and an option can ensure that every marker belongs to at least one annotation.

`make_marker_priors_from_annotations()` constructs marker-level inclusion probabilities and variance multipliers from specified enriched annotations. It uses log inclusion-odds and log variance-multiplier effects, optionally centers the linear predictors, and caps the resulting marker priors.

`sample_causal_pool_weighted()` samples causal markers without replacement, giving markers in selected annotations increased sampling weight.

`mtsim_annotation()` combines annotation generation, weighted causal-marker sampling, and marker-prior construction with multi-trait phenotype simulation. Its returned truth allows direct comparison of simulated architecture and posterior summaries.

`summarize_annotation_signal()` reports annotation size, causal-marker counts, and causal rates. If a fit is supplied, it can also report within-annotation mean PIPs and mean absolute posterior marker effects. Because annotation sets may overlap, these summaries should be read one annotation at a time rather than summed across rows.

10 Overlap, Identifiability, and Double Counting

Overlapping annotations are biologically natural but statistically difficult. If two columns of A identify nearly the same markers, their separate coefficients may be weakly identified even when their joint signal is clear. Rare annotations can produce unstable estimates, whereas very broad annotations may largely reproduce the intercept.

Additive constructions on the logit and log scales provide a coherent way to combine overlap:

$$\operatorname{logit}(\pi_i) = \operatorname{logit}(\pi_0) + A_i\eta_\pi, \qquad \log w_i = A_i\eta_v.$$

However, additive predictors can still create extreme priors when a marker belongs to many enriched sets. Centering, coefficient shrinkage, and explicit bounds are therefore structural safeguards, not merely numerical conveniences.

Interpretation is conditional: an annotation coefficient measures the association remaining after the other included annotations and preprocessing choices are accounted for. Marginal set enrichment and conditional annotation coefficients answer different questions and need not agree.

11 Research-Stage Multi-Trait Extensions

The archived annotation draft also develops extensions beyond the currently exposed fixed, learned, and group CSR wrappers. They are retained here as design directions rather than claims about the public API.

For multiple traits, marker-specific annotation priors can vary by trait:

$$\operatorname{logit}(\pi_{it}) = \operatorname{logit}(\pi_{0t}) + A_i\eta_{\pi,t}, \qquad \log w_{it} = A_i\eta_{v,t}.$$

Annotations could also influence cross-trait sharing. For example, a softmax model could assign marker-specific probabilities to patterns such as null, trait-specific, and shared effects. A further extension could let annotation sets influence the covariance of active multi-trait effects. These models require strong regularization because annotation effects, sharing probabilities, and trait covariance can otherwise be difficult to identify.

The draft additionally proposes warm-started regional multi-trait fits: genome-wide annotation summaries construct priors, and selected regions are then fitted with more detailed multi-trait models. Such a workflow may be computationally useful, but prior reuse, region selection, and uncertainty propagation must be handled explicitly.

12 Regularization, Bounds, and Uncertainty

Annotation-informed priors add flexibility and therefore add failure modes. The following controls address distinct problems:

- **Centering** separates relative marker enrichment from the genome-wide baseline.
- **Standardization** places quantitative annotations on comparable scales.
- **Coefficient shrinkage** limits overfitting in correlated or high-dimensional annotation matrices.
- **Probability bounds** prevent prior inclusion probabilities from collapsing to zero or one.
- **Variance-multiplier bounds** prevent extreme prior scales.
- **Cross-fitting or external training** reduces optimism from empirical-Bayes data reuse.
- **Sensitivity analyses** reveal dependence on bounds, annotation definitions, LD settings, and prior choices.

Posterior uncertainty conditional on a fixed empirical-Bayes prior does not include uncertainty from estimating that prior. This distinction should be made explicit when reporting results.

13 Interpretation and Causal Limits

Annotation coefficients, set enrichments, and marker-specific priors describe how external information interacts with the fitted prior and observed genetic evidence. They do not prove that an annotation or annotated marker is causal.

Apparent enrichment can reflect:

- LD between annotated and causal markers;
- differences in marker density and allele frequency;
- correlated or overlapping annotation definitions;
- phenotype architecture and sample size;
- sparse-LD approximation and marker filtering; and
- model, regularization, and prior-bound choices.

Top markers and enriched annotations should therefore be treated as model-dependent evidence. Comparisons against unannotated models and sensitivity analyses are central to interpretation.

14 Implementation Checklist

Before fitting an annotation-informed model:

- verify marker-row alignment across annotations, sufficient statistics, sparse LD, and posterior outputs;
- inspect annotation column prevalence, ranges, missingness, and correlation;
- reconsider all-zero, all-one, extremely rare, or nearly duplicate columns;
- record intercept, centering, standardization, and binary-column choices;
- record inclusion-probability and variance-multiplier bounds;
- decide whether priors are external, empirically constructed, jointly learned, or group based;
- use cross-fitting or external data when empirical-Bayes reuse is a concern;
- compare prior annotation summaries with posterior annotation summaries; and
- report sensitivity to priors, annotation preprocessing, and LD settings.

15 Related Notes and Sources

- Annotation-informed models gives the main conceptual and practical path through the annotation-aware wrappers.
- Technical SBayesRC-style annotations covers the separate annotation-dependent BayesR mixture prior.
- Workflows and outputs explains compact posterior summaries and the annotation workflow.
- Data representations defines the marker-order and summary-statistics contracts used here.

This appendix preserves and reorganizes the prior-construction derivations from the archived annotation-prior draft. Current wrapper behavior is distinguished from empirical-Bayes and multi-trait designs that remain research-stage extensions.